



[First Semester] [2025-2026]

2. Now you must call these methods in the previous program to complete the lexical analysis of the source code, then you must print the output on the output screen with a message that refer to the type of each token as it explains in the following program.

```
import java.io.*;
import java.util.*;

public class Mainprog {

    public static boolean is_SP(char ch){
        if((ch==' ')|| (ch==';') || (ch=='(') || (ch==')') || (ch=='{' ) || (ch=='}') || (ch=='\n'))
            return(true);
        else return(false); }

    public static boolean is_OP(char ch){
        if((ch=='+' ) || (ch=='-' ) || (ch=='*') || (ch=='/' ) || (ch=='='))
            return(true);
        else return(false); }

    public static boolean is_RW(String tok){
        if(("DEF".equals(tok)) || ("BODY".equals(tok)) || ("int".equals(tok)) ||
        ("byte".equals(tok)) || ("print".equals(tok)))
            return(true);
        else return(false);

    }

    public static boolean is_DIG(String tok){
        int i;
        for (i=0;i<=tok.length()-1;i++){
            if ((tok.charAt(i)>='0')&&(tok.charAt(i)<='9'))
                continue;
            else
                return (false);}
        return (true);

    }

    public static boolean is_ID(String tok){
```

[Language Translators]/ [Third]



int i;
if (tok.charAt(0) >= 'a' && tok.charAt(0) <= 'z') ||
(tok.charAt(0) >= 'A' && tok.charAt(0) <= 'Z'))

[Lab 2 & 3]

[08:30 Am]

Instructor Information

Instructor

[Bahera Hani]

[Ruaa Abdullah]

[Ibtihal Abbas]

Email

[ruaa.abdullah@nahrainuniv.edu.iq]

Hours

[4 Hours]

```

        return (false);}
    }
    else
        return (false);
    return (true);
}

public static void main(String[] args) throws Exception{
    Scanner in; /*read from txt file*/
    in = new Scanner(new File("C:\\Users\\khyro\\Desktop\\nt.txt"));
    char ch;
    String line, tk;
    while(in.hasNext()){
        line = in.nextLine(); /*read one line from file*/
        tk = "";
        for(int i=0; i<line.length(); i++){
            ch = line.charAt(i);
            if (ch!=' '){
                if(is_SP(ch)||is_OP(ch)){/* must be send to LUT*/
                    if (!"".equals(tk)){
                        if (is_RW(tk))System.out.println(tk+" it is RW");
                        else if(is_DIG(tk))System.out.println(tk+" it isDIG");
                        else if (is_ID(tk))System.out.println(tk+" it isID");
                        else System.out.println(tk+" it is Error");
                        tk="";
                    }
                    System.out.println(ch+" it is SP or OP");}
                else tk+=ch;
            }
        }
        else if (!"".equals(tk)){
            if (is_RW(tk))System.out.println(tk+" it is RW");
            else if(is_DIG(tk))System.out.println(tk+" it isDIG");
            else if (is_ID(tk))System.out.println(tk+" it is ID");
            else System.out.println(tk+" it is Error");
            tk = "";
        }
    } /*end for loop*/
    if (!"".equals(tk)){
        if (is_RW(tk))System.out.println(tk+" it is RW");
        else if(is_DIG(tk))System.out.println(tk+" it is DIG");
        else if (is_ID(tk))System.out.println(tk+" it is ID");
        else System.out.println(tk+" it is Error");
        tk="";
    }

    }/*end while loop*/

    in.close(); /*close text file*/

```

[Language Translators]/ [Third]



}; /*end main() method*
[First Semester] [2025-2026]

}; /*end Main class*/
[Lab 2 & 3]

[08:30 Am]

4. Instructor Information

Instructor	Email	Hours
DEF [Hera Hail] Int [Ru'aa Abdullah] [Ibtihal Abbas]	[ruaa.abdullah@nahrainuniv.edu.iq]	[4 Hours]

a is id
, is sp
b is id
; is sp
Byte is rw
x is id
, is sp
y is id
; is sp
BODY is rw
{ is sp
a is id
= is op
1 is dig
; is sp